

KONGSBERG BRANNSTASJON, Norway

WMO: 014730

Lat: 59.6244N

Lon: 9.6375E

Elev: 568

StdP: 14.40

Time Zone: 1.00 (EUC)

Period: 04-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-1.7	3.5	-4.2	4.5	0.9	0.5	5.8	6.5	12.4	39.7	10.4	38.0	1.8	270	0.559

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	19.5	81.3	61.6	77.6	60.3	73.8	58.5	64.0	75.2	62.5	72.8	60.9	70.2	5.3	110

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
(p)															
60.6	80.7	66.2	58.7	75.2	64.8	57.0	70.8	63.7	29.5	75.2	28.4	73.0	27.2	70.4	70.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%		2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
Min		Max		Min	Max		Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)		(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
10.0	8.9	7.6	DB	-4.7	86.1	5.6	2.8	-8.7	88.1	-12.0	89.8	-15.1	91.4	-19.1	93.4	
			WB	-3.9	66.9	3.8	2.1	-6.6	68.4	-8.8	69.7	-10.9	70.9	-13.6	72.5	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	42.4	24.2	26.2	32.9	41.9	50.7	57.8	62.3	59.1	52.3	41.8	32.8	25.5
	DBStd	15.18	9.54	8.90	8.29	5.98	7.01	5.19	5.13	4.21	5.04	6.28	7.43	10.65
	HDD50	3899	800	666	531	251	77	3	0	0	34	262	516	759
	HDD65	8307	1265	1086	996	692	447	226	119	189	380	718	965	1224
	CDD50	1121	0	0	1	9	98	236	381	282	104	10	0	0
	CDD65	51	0	0	0	0	2	9	35	5	0	0	0	0
	CDH74	741	0	0	0	0	66	175	428	68	4	0	0	0
	CDH80	137	0	0	0	0	7	34	91	5	0	0	0	0
Wind	WSAvg	3.4	3.2	3.1	3.6	3.7	4.0	4.0	3.6	3.2	3.4	3.1	3.1	3.0
Precipitation	PrecAvg	38.1	3.0	1.9	1.9	2.3	2.6	3.3	3.6	4.1	4.1	4.3	4.2	2.8
	PrecMax	58.8	6.5	5.4	5.4	5.6	7.7	7.7	8.0	8.1	11.3	10.8	18.1	4.5
	PrecMin	29.8	0.5	0.3	0.0	0.3	0.4	0.2	0.6	0.5	1.8	1.0	0.6	0.6
	PrecStd	5.9	1.6	1.1	1.3	1.3	1.5	1.7	1.7	1.8	2.2	2.3	3.3	1.2
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	46.4	49.1	60.9	68.3	80.5	84.6	86.4	80.1	73.2	62.2	53.5	50.1
		MCWB	40.5	40.6	46.6	51.6	58.7	60.6	63.7	62.1	59.6	54.9	47.5	44.6
	2%	DB	43.0	44.5	54.4	64.2	74.8	79.2	82.5	75.4	68.6	58.2	50.3	45.1
		MCWB	38.3	37.9	43.5	49.9	56.6	59.0	62.9	60.8	56.6	50.9	46.3	40.8
	5%	DB	39.7	40.9	49.2	59.2	70.4	74.6	79.1	72.1	65.6	55.1	46.6	42.2
		MCWB	35.7	36.0	40.1	46.9	54.4	56.8	61.9	59.2	55.1	49.2	43.4	38.4
	10%	DB	36.6	37.6	45.1	54.8	65.5	70.3	75.0	69.0	62.4	52.2	43.6	39.2
		MCWB	33.6	34.2	37.5	44.0	52.6	55.3	60.0	57.9	53.6	47.6	40.9	36.0
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	42.0	41.9	48.1	53.8	61.7	63.1	66.8	64.8	61.6	55.8	49.7	45.2
		MCDB	45.6	47.8	60.1	65.1	74.1	76.2	78.0	73.6	69.4	60.4	51.8	49.5
	2%	WB	39.2	38.9	44.2	50.5	59.2	61.0	64.8	63.1	58.9	52.6	46.6	41.6
		MCDB	42.7	42.9	53.3	61.2	70.0	73.3	77.1	70.9	64.6	56.3	49.1	44.6
	5%	WB	36.6	36.7	41.1	48.0	56.4	59.1	63.4	61.5	56.9	50.6	44.0	38.8
		MCDB	39.2	40.3	48.1	57.6	66.6	71.0	75.2	68.9	62.6	53.7	46.5	41.7
	10%	WB	34.1	34.6	38.4	45.6	53.7	57.2	61.8	59.9	55.3	48.4	41.1	36.2
		MCDB	36.4	37.3	44.0	53.6	63.0	67.1	72.4	66.5	60.3	51.9	43.2	38.8
Mean Daily Temperature Range	5% DB	MDBR	10.7	13.0	18.0	20.1	20.4	19.9	19.5	18.1	17.7	13.8	10.7	11.0
		MCDBR	13.5	17.4	23.8	28.0	28.3	28.2	26.5	23.2	22.9	17.9	15.7	14.3
		MCWBR	10.6	12.1	14.4	15.1	13.0	12.7	11.5	11.2	12.9	11.7	12.4	11.4
	5% WB	MCDBR	12.6	15.2	22.7	25.5	22.7	23.3	22.8	18.8	18.7	14.8	14.3	13.5
		MCWBR	10.6	11.1	14.2	14.2	11.4	11.6	10.5	10.1	11.6	10.8	12.1	11.3
Clear-Sky Solar Irradiance	taub		0.251	0.291	0.317	0.360	0.358	0.348	0.368	0.362	0.331	0.317	0.272	0.226
	taud		2.213	2.291	2.319	2.323	2.425	2.474	2.434	2.446	2.523	2.482	2.308	2.129
	Ebn at Noon		176	230	259	266	276	281	272	264	256	221	168	139
	Edh at Noon		13	21	28	33	32	31	32	30	23	18	12	9
All-Sky Solar Radiation	RadAvg		104	312	795	1267	1541	1758	1541	1220	800	371	136	69
	RadStd		21	65	70	179	186	143	202	93	110	71	35	7

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (53 neighbors)	N/A	N/A	+2.68	N/A	N/A	N/A	N/A	N/A	N/A	N/A